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Roll No

EE/EX-7003-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Computer Application to Power System

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) What are the capability curves of an alternator? Discuss their role in voltage stability of the power system. 7
b) Describe the procedure in steps for the formation of Z bus matrix. 7
2. a) What is the effect of series and shunt compensation in power system? 7
b) Write short notes on the following: 7
i) SVC
ii) SVS
3. a) Discuss various sensitivity analysis techniques. Derive general sensitivity relation of n bus power system. 7
b) Explain the terms: generation shift distribution factor, line outage distribution factor and compensated shift factor in sensitivity. 7

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4. a) Explain the following: 7
 - i) Pre and Post Contingency
 - ii) Corrective load rescheduling
- b) Discuss the method of economic dispatch using LP formulation. 7
5. a) Explain the concept of voltage stability. Differentiate between voltage and angle stability. 7
- b) Discuss the effect of series and shunt compensation on voltage stability. 7
6. a) Derive the elements of Jacobian matrix of a power system. 7
- b) Explain the effect of load models on voltage stability. 7
7. a) Derive the following sensitivity factors 7
 - i) GSDF
 - ii) LODF
- b) Explain V-Q sensitivity 7
8. Write short notes (any two) 2×7=14
 - a) Regulating Transformer
 - b) Corrective Rescheduling
 - c) Line Loadability
 - d) Security function

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